



# INTERMEDIATE

NINETY-DAY PROTOCOL

You know what you should be doing and you do some of it. This turns sometimes into every day.

## Trillions of batteries, keeping time together

Your body is not one battery. It is a coordinated network of trillions of cellular batteries. Every cell holds a voltage across a membrane, and different tissues are built to hold different voltages. A neuron, a muscle cell and an immune cell are supposed to be different. The goal is the voltage appropriate to each tissue, never the maximum.

**Coherence** is the ability of cells to hold their proper electrical identity while coordinating their voltage, metabolism, timing and behaviour with the tissue around them and with the whole organism. A cell can be individually healthy and still be out of step with its neighbours. That is what the ninety days are for.

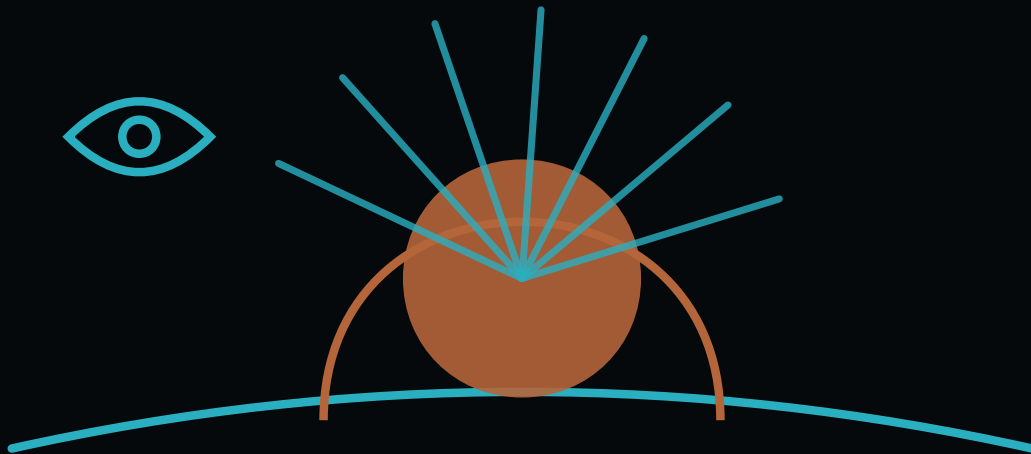
Light does two jobs. The big one is timing. Morning light tells the body what hour it is, and that signal sets the nervous system, hormones, metabolism and mitochondrial activity. The small one is chemistry in your skin: sunlight makes vitamin D and releases nitric oxide, which widens your arteries. So light is a real energy input, and a small one. Your skin takes in a lot of sunlight and nearly all of it becomes heat, because the body cannot turn light into fuel.

Eating with the season is in the protocol because different foods give your body different fuel, and daylength sets how your body handles fuel. Lining those up is the idea being tested. It is not a settled fact, and food does not carry any message about the month it grew in.

The circadian protocol is how coherence is supported. The daily log is how it is tracked. Every element in this document carries a note saying which of the four questions it answers, and where an element answers none of them, it says so.

# Circadian protocol

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.



LIGHT SETS THE CLOCK

1

## Circadian protocol

INTERMEDIATE

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.

Your body runs on a clock, and the clock is set by light. Morning light tells every cell what time it is. Darkness at night lets the repair cycle run. When the clock is right, food is handled correctly, hormones release on schedule, and sleep does its job. When the clock is wrong, nothing downstream works, no matter how well you eat or train.

The sequence, every day:

LIGHT → WATER → MOVEMENT → FOOD → MOVEMENT → LIGHT → DARKNESS

Go outside as soon as you wake up

**What to do:** Outside within 30 minutes of waking. Fifteen minutes minimum, twenty when cloudy. Barefoot on the ground when you can. Face east. No sunglasses.

**Why:** Your body has a clock inside it. The clock does not know what time it is until it sees the sun. When morning light hits your eyes, it tells every cell in your body that the day has started. That one signal sets up your energy, your hunger, your hormones and your sleep for the whole day. If you skip it, your body runs on the wrong time all day, like a clock that was never set.

Barefoot if you can. This is part of the morning light block, not separate from it. We include it because it costs nothing, and because standing on grass keeps people outside for the full time when shoes and a doormat do not. We are not measuring it, and we are not claiming it does anything on its own.

| **Answers Q3, Q4.** Sets the circadian clock every tissue runs on. Sleep timing and light exposure are logged daily.

Get sun on your skin in the afternoon

**What to do:** Fifteen minutes of sun on your arms and face in the afternoon. Build up slowly if your skin is fair.

**Why:** Morning light sets your clock. Midday sun holds it there. It is also the only time of day your skin can make vitamin D, and sun on bare skin widens your blood vessels, which lowers blood pressure. We measure both at day 0 and day 90.

| **Answers Q3, Q4.** Holds the clock set. Vitamin D is on the panel and blood pressure is measured at both ends.

Watch the sun go down

**What to do:** Outside for the last fifteen minutes of light.

**Why:** The light at sunset is red and orange. That color tells your body the day is ending and it is time to start making melatonin, the hormone that makes you sleepy. Seeing the sunset is how your body knows to begin winding down. Skip it and your body does not get the message, and then you are wide awake at midnight wondering why.

| **Answers Q3.** Red light starts melatonin onset, the handover from day signalling to night signalling.

Put on your glasses when the sun goes down

**What to do:** Amber lenses from sunset. Bedroom and living room switched to incandescent or red bulbs. Iris on every screen. Film on the TV.

**Why:** After sunset, your body expects darkness. Light bulbs and screens give off a blue light that looks like daytime to your brain. Your brain sees it and stops making melatonin, and that is why you cannot fall asleep even when you are tired.

Amber or red glasses block the blue. So do warm bulbs and screen filters. It is not about the brightness. It is about the color.

| **Answers Q3.** Protects that melatonin signal from blue light that would tell the body it is still daytime.

Make the bedroom dark and quiet

**What to do:** Phone out of the bedroom. Router on a timer, off 10pm to 6am.

**Why:** Your body does its repair work while you sleep, and it can only do it in the dark. Even a small light, like a phone screen or a charger glow, can be enough to keep the repair from starting. The phone also keeps your brain on alert, waiting for a buzz. Out of the room means your body can finally switch off.

| **Answers Q3, Q4.** Darkness is when repair signalling runs. Sleep timing and regularity are logged daily.

Give your body hours with no food

**What to do:** Eating window 8am to 5pm. A 15-hour fast every night. One 24-hour fast per month once the window is easy.

**Why:** Every time you eat, your body has to stop what it is doing and deal with the food. If you eat from the minute you wake up until right before bed, your body never gets a break to do repairs. Hours without food, especially overnight, is when the cleanup happens. Late eating is the worst, because your body is trying to wind down and you just handed it a job.

| **Answers Q1, Q4.** Metabolic flexibility and substrate switching. Fasting glucose, insulin and HbA1c are on the panel.

**The rule that matters most:** light before food, every single day. Do not eat until you have been outside.

# Water and minerals

The morning glass first. Then three liters, every bottle with minerals in it.



MINERALS, LEMON, SALT

## Water and minerals

INTERMEDIATE

### The morning glass

Before coffee, before food, before anything: 16 to 24 oz of mineral water with a pinch of Baja Gold sea salt and 1 to 2 oz of pure organic lemon juice. Empty stomach. It is the first thing the body receives after light.

### Through the day

Three liters minimum, more on training days. Every bottle gets a pinch of Baja Gold, because water without minerals passes through and takes minerals with it. Most of it before your last meal.

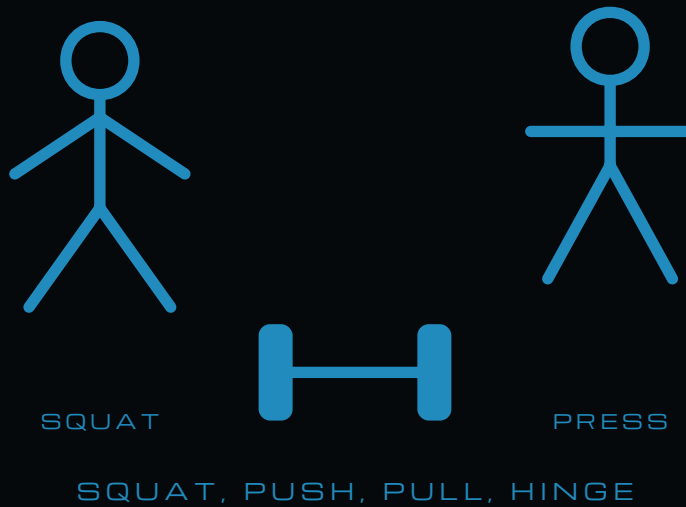
### What to drink

Crazy Water #4, Saratoga or Icelandic Glacial as bottled water. At home, reverse osmosis with Baja Gold added, a quarter teaspoon per liter. Baja Gold for cooking. Nothing else.

| **Answers Q2, Q4.** Minerals are the ions the pumps move. Sodium, potassium and RBC magnesium are on the panel.

# Movement

Four patterns. Walks after every meal. Zone 2 for the mitochondria, intervals for the ceiling.





## Movement

INTERMEDIATE

### Resistance, three days

Bodyweight progressions. **Squat:** bodyweight to goblet to Bulgarian split squat. **Push:** incline push-up to floor to dips. **Pull:** band rows to inverted rows to pull-ups. **Hinge:** band hinge to kettlebell Romanian deadlift. Three sets of eight to twelve. Progress when you clear twelve on all three.

### Zone 2, three sessions of 40 minutes

Brisk walking, incline treadmill, easy cycling. You can talk.

### Intervals, once a week

30-20-10: thirty seconds easy, twenty moderate, ten hard, five times through. Rest two minutes. Repeat the block. Twenty minutes total.

### Walking

Ten minutes after every meal, fifteen after dinner. Eight thousand steps.

### Grip

Dead hangs, three sets to failure, twice a week.

| **Answers Q1, Q4.** Builds mitochondrial number and respiratory capacity. VO2max, lactate and grip strength are measured at both ends.

# Food

Protein first, fat for fuel, inside a window that closes early. The full list is in the Dietary Guidelines.



PROTEIN, FAT, THEN CARBS

## Food

INTERMEDIATE

Window: 8am to 5pm

Three meals or two. No snacks.

First meal

Three pastured eggs, or a tin of sardines four days a week. Leafy greens. A forkful of sauerkraut or kimchi. Olive oil or butter.

Midday

6 to 8 oz protein: beef, chicken thigh, salmon, lamb. Two cups vegetables, half cruciferous. Sweet potato or white rice, one fist. Olive oil.

Last meal, by 5pm

6 oz protein. Vegetables. Avocado.

Every day

Eggs or sardines every morning. Greens every day. Fermented food every day. Salmon or sardines four times a week.

Protein

1.6 grams per kilogram.

Out for 90 days

Gluten, seed oils, added sugar, alcohol. Grains limited to white rice and sweet potato. Nightshades stay in.

**Answers Q1, Q2, Q4.** Fuel handling, plus the fatty acids and protein the membranes are built from. Glucose, insulin, HbA1c, Omega-3 Index and albumin are on the panel.

The full approved food list, the daily non-negotiables and how to build a plate are in the Dietary Guidelines document.

# Heat and cold

Stress that builds capacity. Dosed to your tier, timed away from your lifts.



HEAT



COLD

STRESS THAT BUILDS CAPACITY

## Heat and cold

INTERMEDIATE

Sauna, three days if you have access

Gym sauna is fine. Fifteen to twenty minutes.

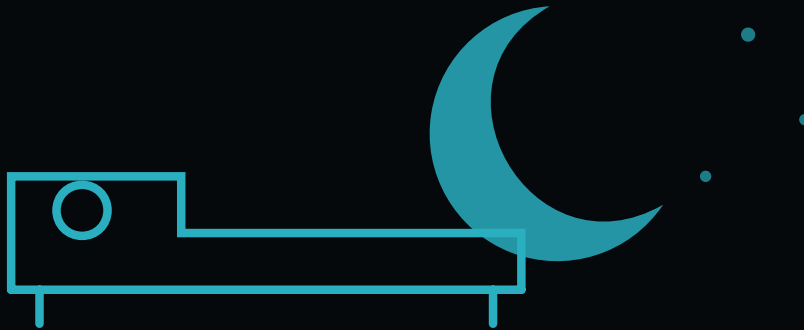
Cold

Every shower ends with sixty to ninety seconds cold. Full cold. Breathe slowly through your nose. That is the entire cold protocol until day 60. After day 60, add one cold plunge a week if you have access.

| **Answers Q1.** Controlled stress that builds mitochondrial capacity. Nothing on our panel isolates it.

# Sleep

Same time, every night. Dark, cool, nothing plugged in.



DARK, COOL, SAME TIME

## Sleep

INTERMEDIATE

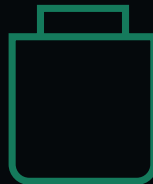
Same bedtime, same wake time, seven days a week. Room cool and completely dark. Last food three hours before bed. Phone out of the bedroom.

Fixed wind-down at 9pm: glasses on, screens off, magnesium taken.

| **Answers Q3, Q4.** Recovery is when gradients are rebuilt and membranes repaired. Sleep timing and regularity are logged daily.

# Supplements

Dosed to your labs, not to a label. No commission on any of it.



DOSED TO YOUR LABS



## Supplements

INTERMEDIATE

### Morning

Pure Encapsulations O.N.E. Multivitamin, 1. Life Extension Super Omega-3, 1,500 mg EPA+DHA if starting under 5%, 1,000 mg above. Vitamin D3/K2, 5,000 IU, adjusted after labs. Creatine, 5 g. Gaia Daytime HPA, 2. Spirulina and chlorella blend powder, 1 tablespoon. Cowboy Colostrum, 1 scoop on waking.

### Afternoon

Dragon Herbs Spring Dragon Longevity Tea. Vital Proteins Collagen, 15 g.

### Evening

Magnesium Glycinate, 300 mg. Gaia Nighttime HPA, 2. Dragon Herbs Lights Out as needed.

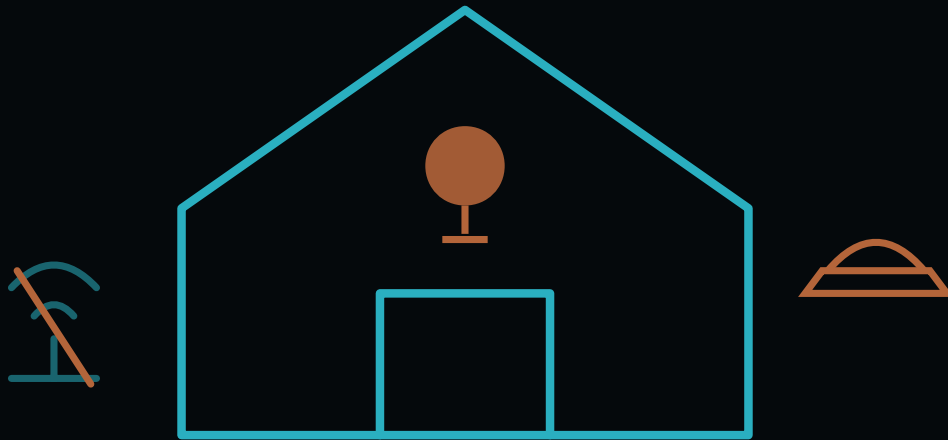
### If tired: the Jing stack

Dragon Herbs Super Adaptogen, 3 morning. Gaia Daytime HPA, 2 morning and 2 afternoon. Cordyceps, 2 before training.

**Answers Q2, Q4.** Substrate for membranes and enzymes. Omega-3 Index, 25-OH vitamin D, RBC magnesium, B12 and ferritin are on the panel.

# Equipment and sourcing

Glasses, bulbs, a router timer, water, and where to get each one.



RED LIGHT, ROUTER OFF, GLASSES ON

## Equipment and sourcing

### INTERMEDIATE

Dr. Pittman takes no commission on anything listed here. These are what he uses.

|                     |                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evening glasses     | Ra Optics Sunset (amber) or Night Shift (red). BLUblox Sleep+. TrueDark Twilight. They must block 460 to 490 nm. Ask for the transmission spectrum.                                                                                                                                                                                             |
| Screens             | Iris (iristech.co) on every computer, Health mode after sunset. f.lux is the free alternative. BlockBlueLight or Ocushield film on monitors. BlockBlueLight filter on the TV.                                                                                                                                                                   |
| Bulbs               | Incandescent wherever you can find them. Otherwise BlockBlueLight or Bon Charge red bulbs for evening rooms.                                                                                                                                                                                                                                    |
| Router              | Mechanical outlet timer, off 10pm to 6am. About twelve dollars.                                                                                                                                                                                                                                                                                 |
| Grounding, optional | <b>Dr. Micah's idea, being tested.</b> Grounding is not a required part of the protocol and the evidence for it is weak. If you want to try it, a conductive sheet for the bed or a mat for the desk is the usual way. Test your outlet with a three-dollar tester first. We name no brand and we take no commission on anything in this guide. |
| Water               | Crazy Water #4, Saratoga, Icelandic Glacial. APEC or iSpring under-sink RO at home. Baja Gold sea salt.                                                                                                                                                                                                                                         |
| Sauna               | Gym sauna for most. Sun Home or Almost Heaven barrel for a home unit. Traditional, not infrared.                                                                                                                                                                                                                                                |
| Cold                | Cold shower to start. Ice Barrel or a chest freezer conversion with a GFCI for a home plunge.                                                                                                                                                                                                                                                   |
| Testing             | OmegaQuant Omega-3 Index kit, day 0 and day 90. Camry grip dynamometer.                                                                                                                                                                                                                                                                         |

# Three phases

Stop the drain. Recharge. Build. Measured at both ends.



MEASURED AT BOTH ENDS

## Three phases

### INTERMEDIATE

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| Days 1 to 30  | Stop the drain. Morning light, fixed wake time, no snacks, last food three hours before bed. Glasses on in the evening. |
| Days 31 to 60 | Recharge. Window to 8am to 5pm. Three resistance days. Cold finish on every shower.                                     |
| Days 61 to 90 | Build. Intervals once a week. First cold plunge. Day 90 draw.                                                           |

10 / EVERY DAY

# Daily checklist

This is what goes in the log. Under a minute.

|                                     |   |                                     |   |                                     |   |
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UNDER A MINUTE, EVERY DAY

10

10 / EVERY DAY

## Daily checklist

INTERMEDIATE

- ☐ Outside within 30 min of waking, 10+ min
- ☐ Morning glass
- ☐ Meals inside 8am to 5pm, no snacks
- ☐ Walk after each meal
- ☐ Midday sun 10 min
- ☐

Training or zone 2 or rest day

- ☐ Cold finish on shower
- ☐ Amber glasses two hours before bed
- ☐ Three liters of water
- ☐ Wind-down at 9pm, phone out of room
- ☐ Logged

## What this is, and is not

The Human Battery Project is an educational wellness program, not medical treatment. It does not diagnose or treat any condition and does not replace your physician.

Any laboratory result outside the reference range is referred to a physician. Every time.

If you take prescription medication, review the supplement list and any change to your eating window with your prescriber before starting. Never adjust a medication on your own because of anything in this document.

Stop and seek medical care for chest pain, fainting, shortness of breath at rest, or any new symptom that concerns you.

Never manipulate potassium or other electrolytes to try to change your body's charge. Balance and correction of a genuine deficiency only.

**We do not promise outcomes.** We measure at both ends and show you exactly what changed.